Memory Banking in 8086

Q. What is Memory Banking in the 8086 microprocessor? Explain the roles of the A0 and BHE#' signals in accessing Even and Odd memory banks for byte and word transfers.

> **Definition to Remember**

> **Me
Bank**
bytes (

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1. Concept of Memory Banking

To support a 16-bit data bus while maintaining byte-level addressing, memory is split:

- **E
D7**).

2. Role of A0 and BHE#' Signals

The CPU uses two specific signals to select which bank to activate during a memory cycle:

- **A

3. Byte and Word Access Scenarios

| BHE#' | A0 | Selection | Transfer Type |

| :---: |

4. Accessing a Word from an Odd Address (Unaligned Access)

If the CPU tries to read a 16-bit word starting at an **odd address** (e.g., `00001H`), it cannot do it in one cycle. It requires **two clock cycles**:

- **C

> **Must-Write Points (for 10 marks)**

> 1. **

> **Quick Recall**

> `Me
cycle